# The Verified Constitutional Kernel

## Machine-Checked Invariants of Axiomatic Humanist Cybernetics v3.1

**Plain Language Summary · Formal Appendix · Reviewer Brief Kernel version 0.12 — 2026-07-13**

Companion artifact to AHC v3.1 and Companion A (integrated). **116 theorems · 4 modules + a cross-module bridge · Lean 4.15.0, core only · zero unproven claims.** Source digest (sha256 over sorted sources): `88ea65837352877ce5d3ad94c78ea244f41b7a1ffc743fce99b172019c490830`

Circulated under Principle 19.2 (Complexity Ceiling): this artifact adds no new specification. It verifies specification that already exists and identifies precisely where verification ends. Every change since the Phase 1 packet followed external review, never preceded it. This version incorporates the third review round: an adversarial report (Reviewer #2, 2026-07-13) found no broken proof but exhibited five behaviors satisfying the Lean model while violating this brief's strongest English claims — two internal composition gaps, two machine-semantics gaps, and a family of overclaiming glosses. The gaps were closed in the kernel (v0.8–v0.10); the glosses were narrowed to exactly what is proved (v0.11); the full accounting is Part II.J, the author response is `external-reviews/2026-07-13-reviewer2-response.md`, and the change record is `ERRATA_AND_AMENDMENTS.md` with every superseded digest retained for audit.

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## Part I — Plain Language Summary

This summary follows the framework's own Plain Language Output Layer requirement (AHC §10.4; Companion A Appendix B), applied to this artifact. A constitutional architecture that requires plain-language accountability of every output owes the same of its own proofs.

### What this is, and what has been done

The AHC constitution contains rules written in careful English: rules about when the system may act and how hard, about how long an emergency may last, about how measurements are combined, about what the public record of a decision must contain. We translated four families of those rules into a language a computer can check with complete rigor — a proof assistant called Lean — and proved 116 statements about them. A proof assistant does not test a rule on examples; it verifies that no counterexample can exist, anywhere, ever, within the rules as written.

This is the twelfth iteration of the kernel and the third to result from external review. The most recent review did exactly what review is for: it found no broken proof, but it found two places where two of our own formal objects should have constrained each other and did not, two places where the machine's behavior was weaker than this document's English, and several sentences that claimed more than the mathematics delivered. All five — including the compositionality gap, the review's structural centerpiece — are now closed by new machine-checked theorems. The sentences have been rewritten — in this document and in the source itself — to say exactly what is proved, no more. The four families:

**1. When the system may act, and how hard (the evidence ladder and the emergency clock).** The system can never act more forcefully than its evidence permits, and everything it does on less-than-conclusive evidence can be undone. An emergency protective order that is not confirmed within 72 hours reverses itself automatically — and the 72-hour limit bounds the *evidence*, not merely the paperwork: a system cannot let a protective order lapse and re-file the same stale case to buy another 72 hours. If real danger genuinely persists past the deadline, the system does not simply lose all protection (that would be inhumane) and does not silently keep its full emergency powers (that would be lawless). It enters a narrow **continuity-hold**: a floor of the mildest, certified-reversible measures, while a mandatory human review decides what happens next. As of this version that review has teeth in both directions. The review's outcome is *typed*: a denial or a close-out can never quietly reset the system so a stale case can be re-filed — only an explicit, publicly disclosed authorization of a new episode reopens the full clock, and genuinely new evidence remains the other path. And the review itself cannot be waited out: a hold left unreviewed for 72 hours under continuing danger lands in a flagged **overdue** state — the protective floor stays (a review breach is never a protection vacuum), but full emergency authority is unavailable until the owed review actually happens, and the record can never disguise an over-deadline hold as an ordinary one. *Continuity of a signal is never continuity of authority.* This is proved.

**2. What "irreversible" means, and when the gravest tools are unlocked.** The system's strongest enforcement mechanisms — cutting off capital, severing a community from part of the network — are not presumed harmless. Whether a given action is reversible is decided per deployment, by a public, contestable certificate that must account for how concentrated it is, how long it lasts, how critical the dependency, how fast it can be rolled back, and what cascades it could trigger. Reversibility is judged by whether the *affected people and institutions* can be restored — not by whether a wire can be reconnected. Without such a certificate, an action is treated as irreversible and requires the highest evidentiary tier (full causal proof). As of this version there is no back door through the emergency system: an emergency order and the continuity-hold floor can authorize an action *only through that same certificate* — the reviewer showed that the older, mechanism-level emergency rules could be read as bypassing the certificates, and that bypass is now provably closed, with the old rules demoted to an outer bound the certified layer can never exceed. A deployment in which *no* amount of a given action is presumed safe — think payroll, medication supply, settlement infrastructure — is a legal configuration. The absence of a certificate is never evidence of reversibility.

**3. How measurements survive corruption.** If a minority of the system's sensors are captured or corrupted, they cannot drag its robust statistics outside the range of honest measurements, and they can never assemble a certified reading entirely on their own. The emergency alarm is derived from these measurements, and here is exactly what that buys — no more: every certified alarm is vouched for by *at least one* honest sensor, so a captured minority alone can neither manufacture the alarm that keeps a community under protective authority nor sustain it. It does **not** yet mean that most honest sensors agree with the alarm: a corrupted coalition plus a single honest outlier can certify one. And the danger threshold the alarm is measured against is, in the current model, an input the deployment must bind to a fixed, public configuration — a system free to move its own threshold could raise the alarm without touching any sensor. Both strengthenings — an alarm derived from the robust median rather than a bare quorum, and a threshold pinned to an attested policy — are named candidates for the next formalization round, not claims of this one.

**4. What the public record of a decision must contain, and when.** If any one community's viability collapses, the system's global health signal reads zero — no prosperity elsewhere can mask it (and we prove that a *sum* would allow masking where the chosen *product* does not; the design choice is what prevents it). The plain-language record and the technical record of any decision cannot disagree about anything *marked as mattering* for challenging that decision while both comply with the rules — and if they do disagree, the disagreement itself proves which record broke the rules. Whether everything that genuinely matters is so marked is a human obligation at the seam, and we now say so explicitly rather than letting "harmless" imply it. Everything needed to challenge a decision is public at the moment the decision is made. For an emergency order the record must state, at hour zero: *why* it was authorized, *what makes any "new evidence" claim genuinely new* (challengeable in itself), *how many hours of unreviewed protection the incident has already consumed*, and *what evidence would prove the order wrong*. As of this version those four fields are typed roles no single sentence can fill twice, and the disclosed hour-count is not merely accompanied by a proof — it *is* the machine's own count, one and the same object, in every version of the record: the immediate one and the later civic and technical ones alike.

## What this system of proofs can and cannot see

The proofs verify the rules as written — not the world the rules act on. They cannot see whether the sensors are truly independent, whether measurements track real harm, whether a reversibility certificate honestly describes reality, whether the recorded history of an incident is its true history, whether the danger threshold has been honestly pinned, whether everything material has been marked contestation-critical, or whether a plain-language document genuinely means what its formal record claims. Those gaps are deliberate and named in the artifact as **the seam**: machines check everything that does not require human judgment, so that human judgment — exercised by affected communities through Layer 0 — is spent only where nothing can replace it.

## What this artifact does not do right now

It does not make the AHC system exist, does not validate its statistical methods, and does not certify any deployment. It certifies that the constitutional text, in the parts formalized, cannot be lawyered into its opposite — and that the resolutions of the questions raised against earlier versions, including the most adversarial round to date, genuinely mean what they say.

## What would prove this wrong, and how to challenge it

Every claim above is checkable by anyone, without trusting us. The complete source is included; one command (`lake build`) re-verifies every proof from scratch in seconds on an ordinary computer. If any proof were broken or missing, the build would fail and say so. The honest points of challenge are not the proofs but the modeling choices — the places where English was turned into mathematics — and the standing question of whether the certificate, disposition, and continuity-hold designs place their trust in the right hands. Each module lists its own choices on its first page; Part III asks reviewers to attack exactly those. The previous review round demonstrated the method works: its findings were exhibited against the model, ruled on, and closed with new theorems inside a version cycle.

If you find a problem — with a proof, with a translation choice, or with this document itself — Part III says exactly where it goes and who must answer it. A broken proof is fixed in the kernel; a wrong translation goes to the constitution's own amendment process; a problem with how burden is placed on communities goes to the Open Problems Register. Disagreement does not need our permission to be filed.

### Where the full technical record is

The complete machine-checked source accompanies this document (`ahc-verified-kernel/`), with build instructions in its README and the theorem-to-specification map in Part II below. This summary is a navigation entry point, not a substitute for that record.

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## Part II — Formal Appendix

### A. Scope and claim

The kernel formalizes the order-theoretic and counting-theoretic core of AHC v3.1: enforcement gating and the evidence–action ladder (§5.4, §9.2–9.3), emergency time and the crisis cap (Companion A §12.3, AHC §10.3, §4.2), measurement and aggregation (§8.3, §4.1–4.2, §1.1), and output-register invariance (§10.4; Companion A §5, §19.4, Appendix B). The claim is exact and limited: the specification, as modeled, provably possesses the 116 properties below. **The proofs verify the specification, not the world.**

**Toolchain:** Lean 4.15.0, core library only — no Mathlib, no external dependencies. **Zero `sorry` placeholders.** No theorem uses classical choice; the total axiom footprint is `propext` and `Quot.sound` (the standard Lean kernel axioms), and **49 of 116 theorems depend on no axioms at all**. `decCompliant` is a verified `Decidable` instance, checked by elaboration and additional to the 116.

### B. Module 1 — Tiered Evidence-Action Protocol (`TieredProtocol.lean`)

**Phase 1 core (§5.4, §9.2–9.3).**

| Theorem | Verified guarantee |
|---|---|
| `tier_monotone` (T1) | Authorization is upward-closed: stronger evidence never removes an authorized response. |
| `severity_le_evidence` (T2) | Response severity never exceeds evidence rank: force cannot outrun proof. |
| `sub_causal_reversible` (T3) | Everything authorized below Tier 3 is reversible. |
| `irreversible_iff_causal` (T4) | Irreversible mechanisms are authorized exactly at full causal identification. |
| `broadcast_universal` (T5) | The transparency broadcast is authorized at every tier: the system can always say what it sees. |
| `pio_ceiling` / `pio_reversible` (T6, T7) | No PIO state authorizes above Tier-1 severity; everything a PIO authorizes is reversible — at MECHANISM granularity, per the Phase 1 table. Since v0.9 this layer is quarantined: deployment-facing authorization is W10–W18. |
| `pio_resolves` (T8) | |

| Theorem | Verified guarantee |
|---|---|
| | A PIO cannot remain pending past 72 hours: confirmed or auto-reversed (state-machine termination). |
| `no_chatter` (T9) | The hysteresis gap makes simultaneous escalate/de-escalate conditions unsatisfiable (instantaneous). |

**Temporal hysteresis — S1–S3 (§9.2; adopted v0.2, finding AHC-P1-002).** T9 holds at an instant but is satisfied even by a zero-width band. S1–S3 state the *temporal* content of the gap: `oscillation_travel` (S1), `flips_travel_anchored` (S2), and `chatter_requires_travel` (S3, headline) — over any trajectory, (flips − 1)·(gap+1) ≤ total variation, so sub-band noise can flip the enforcement posture at most once. Cycling requires genuine, repeated, full-band swings.

**Episode machine, two clocks — E1–E15 (§5.4; adopted v0.4, rulings D-R1A + D-R4; revised v0.10, findings R2-03/R2-06).** The hazard clock (danger persists) and the epistemic clock (evidence advances) are separated. Layer 0 review outputs are typed dispositions (`continueHold | close | newEpisode`); the continuity-hold carries a mandatory-review clock (`holdReviewDeadline = 72`, a flagged deployment constant pending ratification).

| Theorem | Verified guarantee |
|---|---|
| `episode_no_relitigation` (E1) | Across any span with no novel evidence and no NEW-EPISODE disposition — close and continue dispositions may flow freely — *whatever the exceedance and filing pattern*, unconfirmed full protection ≤ 72 hours. Strengthened in v0.10. |
| `episode_single_issuance` (E2) | At most one full-PIO issuance per such span. |
| `expiry_routes_by_risk` (E3) | At the deadline: continuing risk → continuity-hold; subsided risk → spent. No third outcome, no discretion. |
| `reonset_refloors` / `floor_persists` (E4, E5) | The floor follows the risk and is never withdrawn while risk persists unreviewed — the flagged `overdue` state keeps the floor. |
| `hold_resolution_iff` (E6) | Only an explicit new-episode disposition returns the hold to the restart-permitting posture. A denial, a close, or a continue cannot. |
| `novel_restart_from_spent` / `_from_hold` (E7) | An attested materially new claim restarts a full PIO. |
| `exceedance_cannot_restart` (E8) | Without novelty, no input re-enters the full clock from `hold`, `spent`, or `overdue`. |
| `hold_floor_reversible` / `_severity` / `hold_grants_no_more_than_pio` (E9–E11) | The Phase 1 mechanism-level floor bounds (superseded for deployment by W13–W16). |
| `close_cannot_launder` (E12) | A close disposition followed by a stale filing cannot reach the full clock — the third review's two-step restart witness, negated. |
| `hold_clock_bounded` (E13) | |

| Theorem | Verified guarantee |
|---|---|
| | No reachable hold state carries a clock at or past the mandatory-review deadline. |
| `unreviewed_hold_expires` (E14) | Under continuing exceedance with no disposition, the hold reaches the flagged `overdue` state within its deadline and stays there: an infinite *silent* hold is unconstructible. |
| `overdue_absorbing` / `overdue_resolution_iff` (E15) | While risk persists, only a disposition exits `overdue`, and only a new-episode disposition reaches ordinary posture — Module 2's review discipline at PIO scale. |

**Certified reversibility envelopes — W1–W9 (§5.4, §9.3; adopted v0.4, rulings D-R2A + D-R3).** An action carries an opaque deployment descriptor judged by a deployment-certified `Envelope`. Routing *and* severance are irreversibility-bearing unless certified. `cert_tier_monotone` / `cert_severity_le_evidence` (W1, W2), `cert_sub_causal_reversible` (W3), `cert_irreversible_iff_causal` (W4), `uncertified_routing_needs_causal` / `uncertified_severance_needs_causal` (W5 — **Q2 answered: uncertified severance is treated exactly as irreversible**), `certified_route_at_t1` / `certified_severance_at_t2` (W6), `cert_refinement_conservative` (W7), `no_certificate_no_presumption` (W8), `zero_envelope_constructible` (W9).

**Certified emergency authorization — W10–W18 (adopted v0.9, finding R2-01).** The third review's most severe finding: the emergency path still authorized at mechanism granularity, so a routing action outside every envelope was rejected by the certificate layer while remaining authorized by a pending PIO — two authorization APIs, no theorem forcing the stricter one. Closed:

| Theorem | Verified guarantee |
|---|---|
| `pio_cert_ceiling` / `pio_cert_reversible` (W10, W11) | T6/T7 lifted to certified actions: the emergency channel is severity-capped and reversible in the ENVELOPE sense — by certificate, not by the Phase 1 table's declaration. |
| `pio_certificate_backed` (W12) | A PIO authorizes routing only inside its envelope certificate, and no severance and no sanction in any state. |
| `hold_floor_cert_reversible` / `_severity` (W13, W14) | E9/E10 lifted to the envelope-indexed hold policy: floor reversibility is a theorem of the gating, not an asserted field. |
| `hold_floor_certificate_backed` (W15) | The floor's routing carries its certificate; severance and sanctions are unconstructible in it. |
| `hold_cert_grants_no_more_than_pio` (W16) | The certified floor is a remnant of PIO authority under the same envelope. |
| `cert_pio_refines_mech_pio` (W17) | Quarantine: the certified layer never grants an action whose mechanism the Phase 1 table would refuse — the legacy layer survives exactly as an outer presumptive bound. |
| `cert_hold_floor_constructible` (W18) | Non-vacuity: a broadcast-only certified floor exists for every envelope, including the zero envelope. |

**Exposure-indexed certificates and trace safety — W19–W26 (adopted v0.12, finding R2-07).** The third review's structural centerpiece: pointwise envelopes let ten individually certified actions jointly cross a liquidity, dependency, or cascade threshold. Closed with the reviewer's prescribed repair — a state-transition certificate. `TEnvelope` carries an exposure state σ; certification judges each action *at* the accumulated exposure and produces the exposure after it; the constitutive obligation (a structure field) is that certification is only granted where the step keeps exposure inside the demonstrated-reversible region. What σ measures, and whether the region describes reality, are certification questions at the seam; what is proved is that the gating composes.

| Theorem | Verified guarantee |
| --- | --- |
| `trace_tier_monotone` (W19) | T1/W1 at trace altitude: stronger evidence never de-authorizes a trace. |
| `trace_head_certificate_backed` (W20) | Every action of a sub-causal trace is certificate-backed at its own exposure point. |
| `trace_stays_inside` / `_prefix` (W21, W22) | **Headline:** a finite trace authorized below Tier 3 from inside the certified region stays inside it — at the end and at every intermediate point. Joint threshold-crossing by individually certified actions is unconstructible. |
| `pointwise_degenerate` (W23) | Every pointwise envelope is the trivial-exposure special case, authorizing exactly what it did: existing deployments are conserved; the obligation is what was missing, not a new burden. |
| `budget_binds_traces` (W24) | The review's liquidity example, formalized: under the cumulative-budget certificate, TOTAL routed volume over any authorized trace is within budget. |
| `pointwise_admits_joint_crossing` (W25) | The contrast witness (in the style of A1e): two actions each certified at frozen zero exposure whose two-action trace is refused — the old reading's failure mode, exhibited and excluded. |
| `pio_trace_stays_inside` (W26) | Tier-1 (PIO-grade) authorized sequences keep the exposure invariant: the emergency channel cannot accumulate its way across a certified threshold either. |

## C. Module 2 — Crisis Frequency Cap, Structural Review, Axiom II (`CrisisCap.lean`)

**Phase 1 core + composition.** `window_head_bound` (C1), `rolling_window_bound` (C2), `no_permanent_emergency` (C3): the rolling window carries at most T_cap emergency days, at every offset, for any continuation. `no_emergency_in_review` (C4), `review_absorbing` (C5), `review_gate` (C6): no event grants emergency authority during Structural Review, which only a Layer 0 constitutional output can close. G1–G7 (adopted v0.2): the cap arithmetic and the review automaton joined into one daily-step machine with the cap trip *derived* from the window arithmetic.

**`axiomII_dichotomy` (A2a) — stated at its true strength (v0.11, finding R2-09).** Every episode is exactly one of reversibility-claim-valid or locally terminal: the §4.2 classification is exhaustive and exclusive, so "the system corrected itself eventually" and "the harm was irreversible" cannot both be asserted of one episode. This is a *well-formedness lemma* — the linear-order dichotomy on the two latencies. It does not prove that terminality is detected, that iteration halts, or that a terminal finding forces a change of decision procedure; that operational content of Axiom II is a named formalization candidate (Part III, F-4), not a theorem of this kernel.

**Module boundary, stated (finding R2-10).** The composed machine's `requested` input is not bridged to Module 1's evidence-action protocol: the crisis cap bounds the *frequency* of emergency authority; whether a given day was evidence-authorized is Module 1's job, and a bridge theorem tying each `requested` day to a Module 1 authorization is a named candidate. Likewise the cap is generic in `T_cap < W`; a concrete audited instance at the spec's constants (W = 730, T_cap = 180) is a candidate. Neither genericity weakens C1–C3/G1–G7, which hold for every instance.

## D. Module 3 — Byzantine Measurement Consensus, Axiom I (`SensorsAndKernel.lean`)

*Unchanged since v0.1.* `honest_strict_majority` (B1), `exists_honest_le/_ge` (B2), `median_bracketed` (B3): under m > 2f+1, a captured minority cannot drag a median-type statistic outside the honest envelope. `no_corrupt_certificate` (B4): no coalition of ≤ f sensors can certify a reading no honest sensor made. `axiomI_null_kernel` (A1a) through `sum_admits_masking` (A1e): one non-viable subgraph nullifies global viability (intersection and product forms), the global signal is nonzero exactly when every subgraph's is, and aggregation by sum provably admits the Korematsu Illusion that the product rules out.

## E. Module 4 — Plain Language Output Layer: Register Invariance (`PLOL.lean`)

**Phase 1 core.** `hash_bridge_origin` (P1), `tripartite_single_origin` (P1'), `bridge_proves_origin_not_consistency` (P2, the specification's own limitation proved with a witness), `compliant_registers_agree` (P3), `divergence_convicts` (P3'), `no_hostage` (P5), `contestability_at_breach` (P5'), `decCompliant` (P6): compliance is decidable; the divergence audit needs no expertise and no discretion.

**`residual_divergence_noncritical` (P4) and `later_residual_divergence_noncritical` (P13) — renamed to what they prove (v0.11, finding R2-08).** Under compliance, any claim differing between registers is a genuine claim of the event that was *not marked* contestation-critical. The former name ("harmless") and the gloss "framing, never findings" implied more: they are earned only under the additional assumption that the critical classification is COMPLETE — that every material finding is in fact marked critical. That completeness is an extraction / Layer 0 obligation at the seam, now stated as such in the module header. The theorems bound where divergence can live; they do not certify that what lives there is benign.

**D-R4 disclosures — P7–P9 (adopted v0.5) and the typed claim language — P14–P17 (adopted v0.8, findings R2-02/R2-04).** The third review exhibited the gap: the disclosed budget claim was an untyped atom never connected to the attestation whose accuracy P8 proves, and one claim could fill all four mandatory disclosure roles. Closed: `PIOClaim` makes the four roles distinct constructors, and `PIOEvent` carries its `BudgetAttestation` with the critical set required to contain the budget claim bearing *the attested figure*.

| Theorem | Verified guarantee |
| --- | --- |
| `pio_disclosure_at_breach` (P7) | All four D-R4 fields are in the T+0 semantic record at hour zero. |

| Theorem | Verified guarantee |
|---|---|
| `pio_disclosure_divergence_convicts` (P7') | Omission of any D-R4 field from any register proves that register non-compliant. |
| `attested_budget_accurate` (P8) / `attested_budget_bounded` (P9) | A budget attestation cannot misstate the episode machine's accounting, and is ≤ 72h over any novelty-free, authorization-free span (by E1). |
| `pio_roles_distinct` (P14) | The four disclosure roles are pairwise distinct claims, by constructor disjointness — one sentence cannot fill two roles. |
| `pio_register_budget_accurate` (P15) | Every compliant register contains the budget claim whose figure IS the machine's count over the attested history: disclosed = attested = computed, one term. |
| `pio_budget_no_drift` (P16) | Any budget-role claim in the critical set carries the machine's figure: two conflicting critical budget disclosures are unconstructible. |
| `pio_disclosed_budget_bounded` (P17) | The published figure inherits E1's 72-hour bound. |

**Later-register invariance — P10–P13 (adopted v0.7).** `tripartite_critical_consensus` (P10), `later_registers_no_hostage` (P11), `shipped_pio_disclosure_all_registers` (P12), and P13 above extend register invariance across the full release. **Stated precisely (finding R2-11):** `ShippedTripartite` carries later-register compliance as proof obligations — these theorems are postconditions of an externally attested shipped release, not a machine-verified publication workflow; no publication event or release-hour ordering among the later registers is modeled.

## F. Bridge — Exceedance Derivation (`ExceedanceBridge.lean`, Module 1 × Module 3)

The episode machine consumes `exceedance` — the hazard signal that sustains a continuity-hold — as an input. Adopted v0.6, this bridge derives it from the measurement layer: a certified exceedance requires f+1 sensors reporting at or above the danger threshold θ, so by B2 it always carries an honest witness.

**Stated at its true strength (v0.11, finding R2-05).** The property proved is **one-honest-witness non-fabrication**: a captured minority of at most f sensors can neither certify an alarm alone (X2) nor sustain a hold on a fabricated one (X4), and every alarm the machine acts on is vouched for by at least one honest sensor (X1, X3). It is **not** consensus: an f+1 quorum can be met by the corrupt coalition plus a single honest outlier while most honest sensors read below θ. And θ itself is carried per hour, unbound by the kernel across time or policy version — a deployment able to lower θ can raise the alarm without corrupting any sensor. Binding θ to a fixed, versioned, publicly attested configuration is a deployment obligation; deriving exceedance from a robust aggregate (B3 `median_bracketed` is the in-kernel candidate) with a quantified honest-corroboration level is a named strengthening candidate (Part III, F-4).

| Theorem | Verified guarantee |
|---|---|
| `derived_exceedance_honest_witnessed` (X1) | A certified exceedance always has an honest sensor at or above θ: never a pure fabrication of the corrupted coalition. |

| Theorem | Verified guarantee |
|---|---|
| `derived_exceedance_not_forgeable` (X2) | If every honest sensor reads below θ, exceedance cannot be certified. |
| `hold_sustained_only_by_witnessed_danger` (X3) | Whenever the machine keeps a subgraph in a continuity-hold on a derived input, at least one honest sensor witnesses the danger. |
| `manufactured_danger_cannot_sustain_hold` (X4) | If every honest sensor reads below θ, the machine drops the subgraph out of the hold — at any hold-clock value. |

## G. Modeling disclosures — the contestation targets

Per the framework's norm, every translation choice is stated in the header of its module. The most load-bearing:

- **Time is discretized** — hours for the PIO/episode machine, days for the Crisis Cap trace; the 72h PIO constant is the spec's. The hold's mandatory-review period (`holdReviewDeadline = 72`) is set equal to the PIO deadline **pending a constitutional ruling on the constant** — the E12–E15 theorems hold for any positive value.
- **Layer 0 outputs are typed dispositions** (v0.10): continue, close, or new-episode. The *substance* of a disposition and the basis of a new-episode authorization live outside the machine, disclosed through D-R4. From the flagged `overdue` state, novel filings wait on the owed review; Layer 0 can authorize immediately via a new-episode disposition — the same body that attests novelty.
- **The episode's `novel` input is an ATG/Layer 0 attestation** attached to a claim identity (D-R4); `exceedance` is the raw risk signal. The hazard clock and the epistemic clock never share a needle.
- **Reversibility is an opaque deployment certificate** (`Envelope`); the kernel gates on the certificate but does not verify that it describes reality. Since v0.9 the PIO and hold floor authorize only certified actions (`pioAuthorizesC`, `CHoldPolicy`), and W17 proves the legacy mechanism table is an outer bound, not an authorization channel. Since v0.12 certification is exposure-indexed (`TEnvelope`, W19–W26): each action is certified against the deployment's accumulated exposure, with invariant preservation the constitutive obligation. What the exposure state measures is a certification question at the seam.
- **The danger threshold θ is a free per-hour input** the deployment must bind institutionally (finding R2-05); the quorum property is one-honest-witness non-fabrication, not consensus.
- **The budget attestation's figure IS the disclosed claim** (v0.8); whether the *recorded history* is the true history is institutional (append-only logs, Layer 0 audit) — anchoring the history to a canonical log head is a named candidate.
- **Completeness of the critical classification** — that every material finding is marked contestation-critical — is an extraction/Layer 0 obligation at the seam (v0.11, finding R2-08).
- **Constitutive constraints are structure fields** — the hysteresis gap, $T\_cap < W$, the $m > 2f+1$ quorum, the D-R4 roles with the attested figure — so degenerate or withholding parameterizations are unconstructible.

## H. What is not verified, and why

The statistical layer (three-stage causal identification §5.5) is not Lean-shaped; its instrument is pre-registration and blind historical validation (OP.4, Companion B). Viability-kernel dynamics over continuous state spaces are the province of viability theory. Sensor independence is an institutional achievement (OP.1), assumed as the $\leq f$ fault bound. The certificate, disposition, and continuity-hold designs move trust to named institutional points — who issues, contests, and revokes an `Envelope`, `CHoldPolicy`, or Layer 0 disposition; who binds θ; who guarantees an incident's recorded history; who marks the critical set completely — and those points are named, not resolved, here. The relation between a natural-language text and the claim set it expresses — the seam — is formally unverifiable by construction: all trust is consumed at claim extraction, none after it, and extraction belongs to Layer 0. The kernel's contribution is to prove that everything after extraction is mechanical (`decCompliant`), that divergence convicts (`divergence_convicts`, `pio_disclosure_divergence_convicts`), and that the one quantitative disclosure is the machine's own number (P15/P16).

## I. Reproduction

Requirements: any machine with `elan` (the Lean toolchain manager). Then:

```
cd ahc-verified-kernel
lake build
```

The build elaborates every proof and prints the axiom audit (`AHCKernel/Audit.lean`) to the log. A missing or broken proof fails the build; a placeholder surfaces as the `sorryAx` axiom. **Expected:** 116 audited theorems, each at most `[propext, Quot.sound]`; never `Classical.choice`; 49 axiom-free. A continuous-integration workflow re-runs this check on every change and fails on any deviation from the 116/49 footprint.

## J. Disposition of the third review round (Reviewer #2, 2026-07-13)

| Finding | Severity | Disposition |
|---|---|---|
| R2-01 emergency path bypasses the certificate layer | Critical | **Closed in Lean** — v0.9, W10–W18 (quarantine theorem W17). |
| R2-02 disclosed budget claim not linked to its attestation | Critical | **Closed in Lean** — v0.8, P14–P17 (one term by construction). |
| R2-03 Layer 0 Boolean launders a stale restart | High | **Closed in Lean** — v0.10, typed dispositions; E6 revised, E12. |
| R2-04 four disclosure fields may be one atom | High | **Closed in Lean** — v0.8, P14 (constructor disjointness). |
| R2-05 exceedance overclaimed as consensus; θ free | High | **Claims narrowed** (v0.11, Part II.F); θ-binding and median-derived exceedance are named candidates. |
| R2-06 continuity-hold unbounded | High | **Closed in Lean** — v0.10, clocked hold, `overdue`, E13–E15. |

| Finding | Severity | Disposition |
|---|---|---|
| R2-07 envelope predicates not compositional | High | **Closed in Lean** — v0.12, `TEnvelope` and W19–W26, per the report's prescribed repair. |
| R2-08 "harmless" ≠ "not marked critical" | Medium | **Renamed and re-scoped** (v0.11, P4/P13); completeness stated as a seam obligation. |
| R2-09 Axiom II dichotomy near-tautological | Medium | **Re-glossed as well-formedness** (v0.11); operational theorem is a named candidate. |
| R2-10 crisis cap not evidence-bridged; constants generic | Medium | **Boundary stated** (Part II.C); bridge and concrete instance are named candidates. |
| R2-11 shipped-release compliance assumed | Medium | **Stated as postconditions** of an attested shipped release (Part II.E). |
| R2-12 packet not self-contained for faithfulness review | — | **Packaging commitment**: the next circulation packet will include pinned excerpts of AHC v3.1/Companion A with a theorem-to-text map. |
| Documentation items 1–7 | — | Items 1–4 fixed; checksums manifest, provenance attestation, and CI pinning are packaging work in progress. |

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## Part III — Reviewer Brief

Companion A Principle 19.2 holds that further specification must follow, not precede, external review. Three rounds have now done so: Phase 1 findings (AHC-P1-001..008), the constitutional-review findings (CG-1..CG-6) with rulings D-R1A/D-R2A/D-R3/D-R4, and the adversarial third round (R2-01..R2-12) whose disposition is Part II.J. Notably, the third round answered a question this brief's predecessor posed: v0.7's F-2 asked whether the hold should carry "a bound … before mandatory resolution", and R2-06's answer is now theorem E14. This round solicits attack on the repairs themselves, and on the surfaces they expose.

### For formal-methods reviewers

**F-1 (Faithfulness of the repairs).** For each third-round repair, does the formal reading weaken or strengthen the constitution's intent? Specific targets: does `overdue`'s refusal of novel-evidence restarts (a materially new claim must wait on the owed review, or arrive as a new-episode disposition) match §5.4's intent, or over-punish the protected community for the institution's failure to review? Is routing every disposition's substance outside the machine the right seam placement? Does W17's quarantine leave any deployment-facing use of the legacy mechanism table that should be forbidden outright?

**F-2 (Disposition and overdue semantics).** The typed dispositions collapse Layer 0's decision space to three outcomes. Is that the right partition — should a denial be distinguished from a close, or a conditional continuation carry structure? From `overdue`, exceedance subsiding returns the subgraph to `spent`: does the review obligation rightly evaporate with the risk, or should a breach flag survive into the record's ordinary posture?

**F-3 (The exposure-indexed certificate — review of the R2-07 repair).** v0.12 implements the report's prescribed state-transition certificate (`TEnvelope`, W19–W26): certification consumes and produces an exposure state, and invariant preservation is the constitutive obligation. Solicited: is a single per-deployment exposure state the right altitude, or do distinct resources (liquidity, dependency, cascade) need distinct, jointly-constrained states? Should exposure *decay* (rollbacks and restorations reducing it), and what does a sound decay obligation look like? Is `Inv`-preservation the right obligation, or should the certificate carry a quantitative margin? And does anything in D-R2A/D-R3's intent require the trace regime to be mandatory rather than the pointwise degenerate case remaining constructible (W23)?

**F-4 (Strengthening candidates, updated).** Named and open: (i) exceedance from a robust aggregate (B3) with a quantified honest-corroboration level, and θ bound to an attested configuration (R2-05); (ii) the operational content of Axiom II — `locallyTerminal` forcing a prohibited transition or mandatory structural review (R2-09); (iii) a Module 1 × Module 2 bridge tying each crisis-cap `requested` day to an evidence authorization, and a concrete audited instance at W = 730, T_cap = 180 (R2-10); (iv) anchoring the budget attestation's history to a canonical append-only log head (L-2); (v) a publication-event model making P10–P13 unconditional (R2-11).

## For constitutional and domain reviewers

**D-1 (Certificate governance).** Rulings D-R2A/D-R3 place reversibility in a deployment certificate that must be public, contestable, and revocable — and since v0.9 the emergency path cannot act outside it. Who issues it, who may contest it, and what happens to actions taken under a certificate later revoked?

**D-2 (The novelty determination).** D-R4 makes `novel` a contestable Layer 0 attestation. Who adjudicates novelty under time pressure during an ongoing incident, and does the contestability right function before, not only after, the protective window it governs?

**D-3 (Hold humaneness in deployment).** The continuity-hold prevents an abrupt protection vacuum (E5), and now cannot be left silently unreviewed (E14). Does a real deployment's certified floor actually protect an affected community, or could a minimal floor satisfy the theorems while leaving people materially unprotected?

**D-4 (Severance certification in the field).** D-R3's soft-severance conditions are constitutional text. Are they the right conditions, and is "tested restoration rather than theoretical reconnection" operationalizable?

**D-5 (Disposition governance — new this round).** The typed dispositions and the `overdue` state give Layer 0's review real mechanical force: a new-episode disposition is now the only non-evidentiary path to full authority, and an unperformed review parks a community's emergency apparatus in a breach state. Who staffs that review at the required cadence, what happens to communities whose institutions chronically run `overdue`, and is `holdReviewDeadline = 72` the right constant to ratify?

## For community and legibility reviewers

**L-1 (Reflexive compliance).** Part I claims PLOL conformance for this document. Audit it against Appendix B.2 and the D-R4 fields: is anything a community reviewer needs missing, and does the falsification pathway function for a non-expert with the Community Statistical Support Fund's resources?

**L-2 (The budget figure's meaning, updated).** P15/P16 now make the disclosed figure the machine's own count — the attachment gap the third round exhibited is closed. What remains is the deeper half: the figure is only as honest as the recorded incident history. Is "the recorded history is the true history" a burden that quietly falls on communities, and what institutional guarantee (append-only logs, independent Layer 0 audit, the named log-head anchor) does it require?

**L-3 (The seam's placement, revisited).** This cycle moved trust to further named points — disposition substance, θ binding, critical-set completeness — alongside certificate issuance, novelty attestation, and history integrity. Are these the right custodians, or does the accumulation relocate expert burden onto communities across several fronts at once?

## Disposition of findings

Findings against proofs (F-class) invalidate specific theorems and are corrected in the kernel. Findings against modeling choices and rulings (F-1, D-class) are constitutional-interpretation questions and route to the specification's amendment processes; the kernel then re-verifies the amended reading — as it has done through eleven versions and three review rounds. Findings against the seam's placement (L-class) route to the Open Problems Register. In all three cases the artifact's function is the same: it converts disagreement about what the constitution means into a precise, checkable object — which is what a formal layer is for.

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*— End of circulation brief. The machine-checked source, the module READMEs, the theorem-to-specification map, the amendment record (`ERRATA_AND_AMENDMENTS.md`), the author response to the third review (`external-reviews/2026-07-13-reviewer2-response.md`), the archived external reviews, and the CI workflow accompany this document. —*